

CoRoT-2b

R-1014

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_160502 · created 2026-07-30T03:16:17+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457509.1409 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.222 +/- 0.065
Transit depth (Rp/Rs)^2	4.91 +/- 2.89 [%]
Orbital Inclination (inc)	89.02 +/- 2.08
Airmass coefficient 1 (a1)	1.46 +/- 0.18
Airmass coefficient 2 (a2)	-0.125 +/- 0.072
Scatter in the residuals of the lightcurve fit is	11.86 %
Best Comparison Star	#2 - [589, 215]
Optimal Aperture	4.22
Optimal Annulus	4.22
Transit Duration (day)	0.0293 +/- 0.0039

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.222 $\pm$ 0.065 vs 0.1667 (deviation 0.85 σ)
Duration fit vs published	0.0293 d vs 0.095 d (deviation 16.85 σ)
Mid-time O-C	-2511.3 min
Depth SNR	3.4
Residual scatter	11.86 %

Light curve

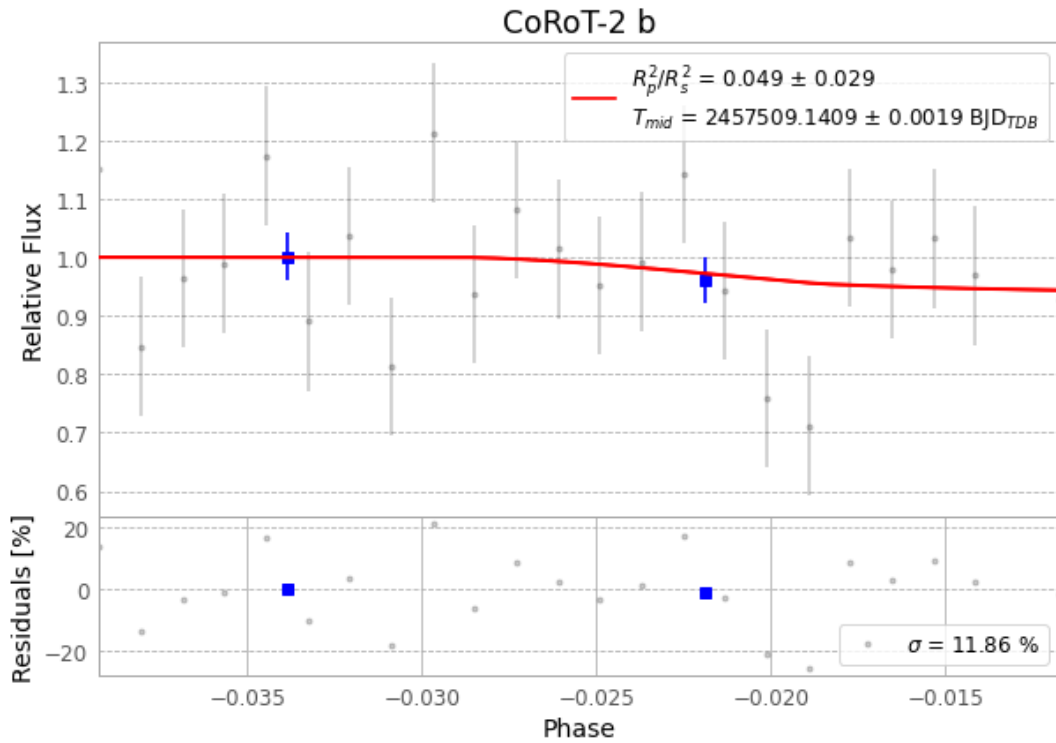


Figure 1 — FinalLightCurve_CoRoT-2 b_2016-05-02.png (SHA-256 070e531d869f7849...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [538, 481], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8993cec616510e22...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

84 FITS frames (55 MB), CoRoT-2160502073012.FITS → CoRoT-2160502113912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-160502.log (88dc883b20e8...), FinalLightCurve_CoRoT-2 b_2016-05-02.png (070e531d869f...), FinalLightCurve_CoRoT-2 b_2016-05-02.csv (67f60b4ffc2b...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 03:16Z.